Abrar Ul Karim

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Basundhara R/A, Dhaka, Bangladesh

CAREER OBJECTIVE

Computer Science and Engineering undergraduate at American International University-Bangladesh (AIUB) with hands-on experience developing web applications and backend systems using React, Next.js, NestJS, ASP.NET Core MVC, and relational databases. Strong interest in software engineering, system design, and machine learning research. Experienced in applying programming and problem-solving skills to academic and personal projects. Seeking a Software Developer opportunity to contribute to practical software solutions while continuing to grow as an engineer.

EDUCATION

American International University-Bangladesh **Dhaka, Bangladesh**
Bachelor of Science (BSc) in Computer Science & Engineering
CGPA: 3.84/4.00 | Expected Graduation: 2027 *Expected 2027*

ACHIEVEMENTS

Dean's List Award — American International University-Bangladesh (AIUB)
Fall 2024–25, Fall 2025–26

PROJECTS

InternNova — Internship Management Platform | Next.js, NestJS, TypeScript, PostgreSQL, TypeORM
GitHub | Live

- Developed a web-based internship platform connecting students, alumni, and companies with role-based access control.
- Implemented company management and internship application modules using NestJS, TypeORM, and DTO validation.
- Integrated JWT-based authentication and role-based authorization for students, HR representatives, and administrators.
- Designed relational database entities and REST APIs for company and application management.

Telecommunication Management System | C#, .NET Windows Forms, MySQL Server
GitHub

- Built a hybrid fraud detection system using XGBoost, CatBoost, CNN, and BiLSTM models.
- Applied meta-learning fusion to combine machine learning and deep learning predictions.
- Cleaned, verified, and validated labeled transaction datasets for model training.
- Added Explainable AI support using SHAP and Integrated Gradients.
- Deployed real-time risk analysis using FastAPI.

TECHNICAL SKILLS

Programming Languages: C++, Java, C#, JavaScript, TypeScript, PHP, Python
Frameworks: React, Next.js, NestJs, ASP.NET Core MVC
Core Areas: Data Structures, Algorithms, OOP, Database Management, Data Science, Machine Learning
Tools: Git, GitHub, Postman
Data/ML: Pandas, NumPy, Jupyter Notebook, XGBoost, CatBoost, SHAP, dataset cleaning, label validation
Database Management: PostgreSQL, SQL Server, MySQL, Oracle Database

PUBLICATIONS

Optimizing Surgery Room Scheduling with Comparative Metaheuristic Algorithms
Accepted at IEEE RAAICON 2025
Compared five metaheuristic algorithms PSO, ACO, GA, SA, and Tabu Search for emergency surgery room scheduling, analyzing execution time, memory usage, cache behavior, and computational efficiency.
DOI:10.1109/RAAICON69033.2025.11502411

EXPERIENCE

Tech Wing's Lab **Dhaka, Bangladesh**
Research Member *May 2025 – Present*

LANGUAGES

English: Fluent, professional working proficiency **Bengali/Bangla:** Native